Trainings

Select Service Tools

Recommended Cummins® Service Tools

- Plugs, Part Numbers 4918322, or equivalent
- Protective Plug Kit, Part Number 4918319, or equivalent
  - Reusable plugs, Part Number 4918322, or equivalent
- Plug, Part Number 4918323, or equivalent
- Plug, Part Number 4918340, or equivalent
- Gasket scraper or cleaning pad, Part Number 3823258, or equivalent
- Scotch-Brite™ 7448 abrasive pad, or equivalent
- Oil Cooler Leak Test Kit, Part Number 4918200, or equivalent
- Threaded plug, Part Number 3089567, or equivalent
- Filter head cap, Part Number 4918204
- Air pressure regulator, Part Number 3164231, or equivalent.

Additional Service Items

- Clean shop towel.

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

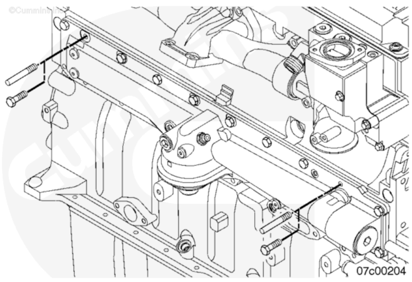
Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries. See equipment manufacturer service information.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/377/377-008-018-tr.html)
- Remove the lubricating oil filter. Refer to Procedure 007-013 in Section 7. (/qs3/pubsys2/xml/en/procedures/377/377-007-013.html)
- Remove the variable geometry turbocharger (VGT). Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/377/377-010-033.html)
- Remove the exhaust gas recirculation (EGR) valve. Refer to Procedure 011-022 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-022.html)
- Remove the EGR cooler connection. Refer to Procedure 011-024 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-024.html)
- Remove the EGR cooler. Refer to Procedure 011-019 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-019.html)

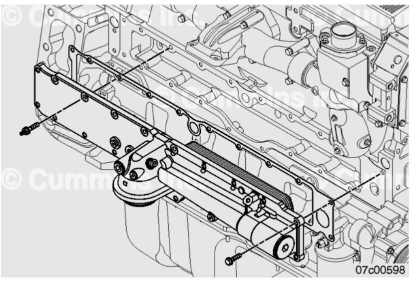
Remove

Remove two oil cooler mounting capscrews and install the guide pins.

Clean the area above the oil cooler to prevent debris from entering the oil passages.

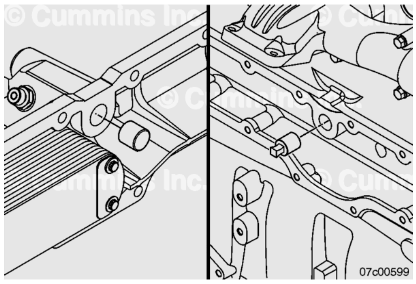


Remove the remaining capscrews, oil cooler, and gasket.
Discard the gasket.



⚠ CAUTION ⚠

Immediately upon removal of the lubricating oil cooler assembly, a plug must be inserted into the housing and cylinder block oil passage drillings. Failure to insert the oil passage plug can result in a bearing malfunction, crankshaft malfunction, or both.

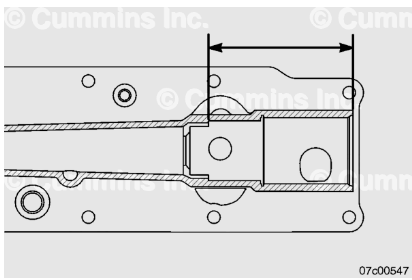


Reusable plugs, Part Numbers 4918322, 4918323, and 4918340, are included in Protective Plug Kit, Part Number 4918319. These plugs are necessary to prevent debris from entering the lubrication system during the repair.

Plug the oil cooler housing filtered oil passage with plug, Part Number 4918323, the block filtered oil passage with plug, Part Number 4918340, and the block oil pump passage with plug, Part Number 4918322.

Firmly push the plugs into the oil passages to prevent all debris from entering the lubrication system when cleaning the gasket surface.

Clean and Inspect for Reuse



Inspect the lubricating oil thermostat seat depth.

Remove the lubricating oil thermostat. Refer to Procedure 007-039 in Section 7. (/qs3/pubsys2/xml/en/procedures/377/377-007-039.html)

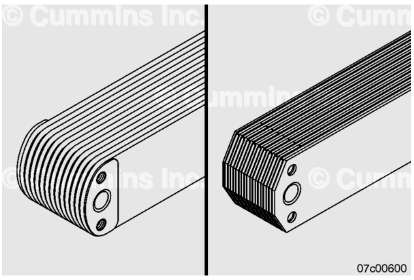
Measure from the spot face to the lubricating oil thermostat seat. Use the measurement points shown in the illustration. If the measured dimension for the lubricating oil cooler housing with integrated filter head is less than the minimum dimension, the oil cooler housing will need to be replaced.

Lubricating Oil Cooler Housing Seat Depth		
mm		in
126.32	MIN	4.973

Install the lubricating oil thermostat. Refer to Procedure 007-039 in Section 7. (/qs3/pubsys2/xml/en/procedures/377/377-007-039.html)

⚠ CAUTION ⚠

Do not reuse an oil cooler core after a debris-related engine malfunction since there is no practical method for cleaning the cooler core. Metal particles which can circulate through the lubricating system can remain in the cooler core and cause engine damage. Do not allow dirt to enter the oil passages when cleaning the oil cooler.

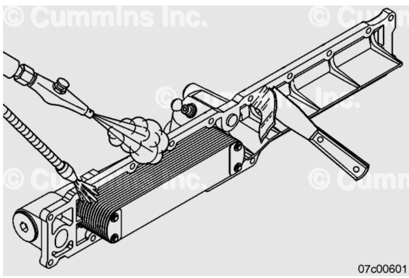


If the inspection requires a new cooler element to be installed, be sure the proper element is installed per the engine family.

- Ratings greater than 450 hp use a cooler with sharp corners that is gold-looking in color.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Do not allow dirt or foreign material to enter oil passages in the cooler housing when cleaning the gasket sealing surfaces. Connecting rod bearing malfunction can be caused if debris is introduced into the cylinder block or lubricating oil cooler housing oil passages.

Plug the oil cooler passages with rubber plugs, Part Numbers 4918322, 4918323, and 4918340, included in kit, Part Number 4918319, to prevent debris from entering the cooler when cleaning the gasket surface.

Clean gasket surfaces by hand with a gasket scraper or cleaning pad, Part Number 3823258. If the gasket material residue can **not** be felt with a fingernail, the surface is ready to accept the new gasket.

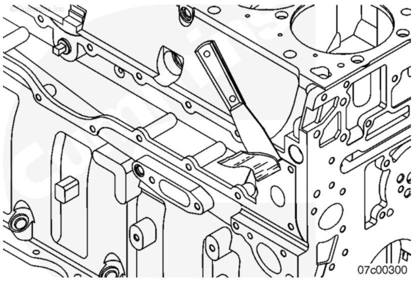
Use solvent to clean and flush the oil cooler passages and cores.

Blow dry with compressed air.

Remove the rubber plugs prior to assembly.

⚠ CAUTION ⚠

Do not allow dirt or foreign material to enter oil passages in the cylinder block when cleaning the gasket sealing surfaces. Connecting rod bearing malfunction can be caused if debris is introduced into the cylinder block or lubricating oil cooler housing oil passages. Therefore, use of power tools combined with abrasive pads to clean gasket surfaces is not recommended.



Plug the oil passages in the cylinder block with rubber plugs.

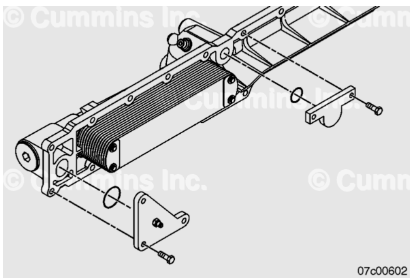
Clean the oil cooler to cylinder block gasket surface with Scotch-Brite™ 7448 abrasive pad.

Remove the rubber plugs prior to assembly.

Leak Test

It is important to leak test the oil cooler assembly as a unit, so leaks at the mounting joints can be identified. These leaks can **not** be found when testing individual elements that have been removed from the assembly. An element is to be replaced **only** if it is found to be leaking.

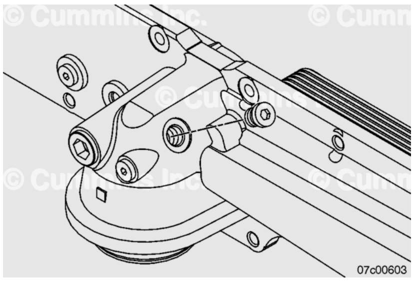
This procedure is **most** effective when the oil cooler assembly is tested in water below 7°C [45°F].



Use the mounting capscrews from the lubricating oil cooler to attach the two plates and corresponding o-rings from Oil Cooler Leak Test Kit, Part Number 4918200, to the lubricating oil cooler housing.

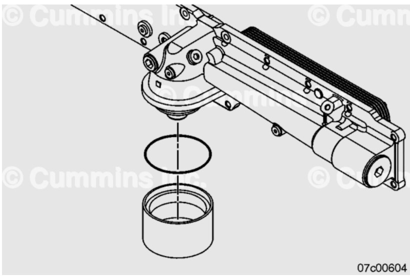
Remove the threaded male union that connects to the turbocharger oil supply.

Install the threaded plug, Part Number 3089567.



Insert the o-ring from the oil filter into the filter head cap, Part Number 4918204.

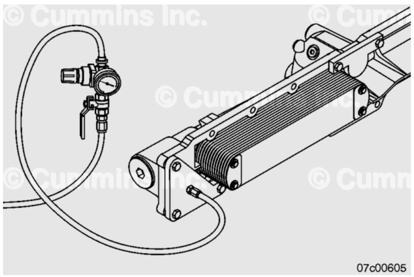
Use engine oil to lubricate the gasket surface and install the filter head cap.



Attach air pressure regulator, Part Number 3164231, to the outlet plate air fitting and pressurize the assembly to 173 kPa [20 psi].

Pressure decay readings **must** be observed three consecutive times to be considered accurate.

Monitor pressure gauge and determine rate of pressure decay for 15 minutes. Allowable pressure drop is 7kPa [1 psi] maximum.

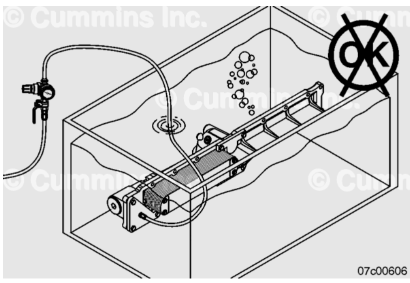


If the lubricating oil cooler does **not** meet the maximum pressure decay requirements from the leak test, perform the following dunk test to identify the probable source of the leak.

Place the oil cooler assembly under water.

If bubbles are observed, carefully determine the source of the bubbles and replace malfunctioning components as necessary.

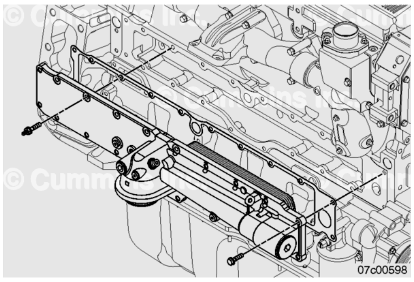
If the element is suspected of leaking, use the following procedure for leak testing individual elements. Refer to Procedure 007-007 in Section 7.



(/qs3/pubsys2/xml/en/procedures/377/377-007-007.html)
Mounting joint leaks are much more common than element leaks. Do **not** replace components that do **not** leak.

Install

Install the guide pins.

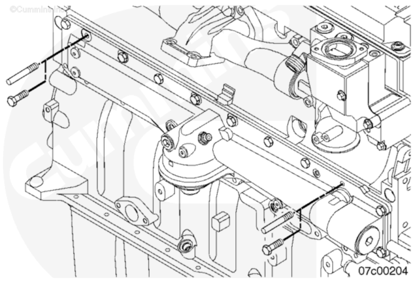


Remove the protective plugs from the oil passages. Make sure no debris enters the lubrication system. Use a clean shop towel to carefully wipe out the oil passages. Inspect to make sure no debris is left in these passages.

Install a new mounting gasket and the oil cooler housing assembly.

Install the capscrews.

Remove the guide pins and install the remaining capscrews.



Tighten the capscrews in a circular pattern from the center out.

Torque Value: 47 n•m [35 ft-lb]

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the EGR cooler. Refer to Procedure 011-019 in Section 19. (/qs3/pubsys2/xml/en/procedures/377/377-011-019.html)
- Install the EGR cooler connections. Refer to Procedure 011-024 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-024.html)
- Install the EGR valve. Refer to Procedure 011-022 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-022.html)

- Install the lubricating oil filter. Refer to Procedure 007-013 in Section 7. (/qs3/pubsys2/xml/en/procedures/377/377-007-013.html)
- Install the VGT. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/377/377-010-033.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/377/377-008-018-tr.html)
- Prime the lubricating oil system. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/377/377-007-037.html)
- Fill the engine with clean oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/377/377-007-037.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to normal operating temperature. Check for oil and coolant leaks.